

Arthur Lin

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EDUCATION

University of Massachusetts Amherst - Manning College of Information and Computer Sciences
B.S. in Computer Science | GPA: 3.88/4

Amherst, MA
Expected May 2028

Relevant Courses: Data Structures & Algorithms, Calculus III, Statistics, Linear Algebra

EXPERIENCE

Gastric Cancer Detection using Deep Learning | Project Leader

Sep 2024 - Present

- Built a hybrid deep-learning CAD system for gastric cancer detection from histopathology images.
- Used EfficientNetV2-B0 and CatBoost to achieve 93.9% accuracy across 10,000+ scans.
- Deployed model via Flask/FastAPI REST API and validated interpretability with Grad-CAM and t-SNE.

University of Southern California | Machine Learning Intern

Jun 2024 - Aug 2024

- Built a semi-supervised ML system using analog VMM on multi-layer memristor crossbars.
- Addressed device variability, I-V nonlinearity, and sneak-path effects using calibration, nonlinear modeling, and SPICE simulation.
- Achieved a 6.6% reduction in standard error through hardware-aware optimization.

MIT Beaver Works CubeSat Challenge | Lead Programmer

Dec 2023 - Apr 2024

- Programmed and tested a microsatellite imaging system for disaster-relief scenarios.
- Used IMU acceleration data to trigger camera capture and automated image transmission to GitHub.

FIRST Tech Challenge, Team #23708 | Team Captain

Sep 2022 - Feb 2024

- Led mechanical design and programming across the geartrain, drivetrain, and claw while coordinating team workflows.

MIT Beaver Works Summer Institute | Selected Member

Jul 2023 - Aug 2023

- Implemented Canny edge detection and Hough transforms for lane-boundary detection in AUV video.
- Generated navigation decisions and validated perception outputs using Matplotlib and AprilTag detection.

MIT Beaver Works CREATE Challenge | Team Captain

Dec 2022 - Apr 2023

- Guided prototyping and testing of an assistive walker with button-controlled braking.
- Received the Best Co-Designer Collaboration Award among 30+ national teams.

SKILLS

Programming Languages: Python, Java, C++

Libraries & Tools: NumPy, Matplotlib, PyTorch, Pandas, Keras, CatBoost, scikit-learn, OpenCV, Flask, FastAPI, Git, SPICE

SELECTED PROGRAMS, AWARDS & HONORS

- Gold Division, United States of America Computing Olympiad (USACO)
- Bronze Medal, President's Volunteer Service Award (PVSA)
- Tutored 10+ students in clear writing and the fundamentals of C++ and Python.